

Jay Gurung

Audio Visual Reactive Tech System

ENT 4501

November 24, 2025

12-Week Project Timeline (WearableTech-AV)

Weeks 1–9: Completed Work

Weeks 10–12: Remaining Work

Week 1 — Ideation & Early Research (Completed)

- Defined core concept: audio-reactive LED wearable device
- Initial research on ESP32, MEMS mics, FFT analysis, LED libraries
- Gathered examples, benchmarks, and design approaches

Week 2 — Proof of Concept Setup (Completed)

- Tested ESP32 with FastLED and basic LED animations
- Connected microphone to confirm raw audio input
- Built very early audio → light reaction sketch
- Confirmed core technical feasibility

Week 3 — Hardware Experiments & Sensor Tests (Completed)

- Tested multiple MEMS mics (INMP441, Fermion)
- Verified I2S pins and stable audio sampling
- Experimented with potentiometer sensitivity mapping
- Early breadboard prototype

Week 4 — First Prototype Structure (Completed)

- Audio-reactive code with FFT bands working
- Stable 5-LED demo version
- Added basic UI elements in p5.js
- Started systematic Agile documentation

Week 5 — Adding System Upgrades (Completed)

- Added additional potentiometers for more control
- Integrated IR receiver
- Introduced multi-mode LED behavior
- Improved code organization and sensitivity tuning

Week 6 — Schematic Design in KiCad (Completed)

- Created complete schematic
- Selected footprints, verified pin mappings
- Added power module, mic, potentiometers, LEDs
- Troubleshoot symbol/footprint mismatches

Week 7 — PCB Layout (Completed)

- Full PCB routing + organization
- Added mounting holes, labels, and silkscreen
- Verified trace width, spacing, and 3D preview
- First 3D file export (.STEP) for enclosure work

Week 8 — Casing Development in Fusion 360 (Completed)

- Imported PCB 3D model into Fusion
- Designed early enclosure structure
- Identified screw-hole size, clearances, and alignment issues
- Tested placement for LED strip & pot cutouts

Week 9 — Repository Audit & Documentation Cleanup (Completed)

- Organized GitHub repo into clear folders
- Added Agile logs, updated README structure
- Documented POC → LVP progression
- Began preparing presentation outline

Weeks 10–12: Future Timeline (Remaining Work)

Week 10 — Final Hardware & Enclosure Completion

Goal: Lock in all physical components

- Finalize PCB (silkscreen, mounting adjustments, cleanup)
- Export/verify Gerbers, drill files
- Finalize Fusion 360 enclosure (mounting pillars, vents, LED slots)
- Test full hardware with updated code
- Re-check all wiring logic and assembly spacing
- Add updated media (photos/screenshots) to repo

Week 11 — Full System Testing & Documentation Build

Goal: Make the system presentation-ready

- Print enclosure test or finalize existing enclosure
- Assemble complete system (ESP32 + mic + pots + LEDs + case)
- Run stability tests (brightness, noise floor, mic gain)
- Polish Arduino & p5.js code
- Finalize README + documentation navigation
- Add diagrams (wiring, PCB screenshots, enclosure images)
- Write Owner's Manual / Process Notes draft

Week 12 — Final Presentation & Polishing

Goal: Deliver final showcase version

- Record final demo video (multiple modes + performance demo)
- Complete slideshow (problem, process, challenges, result)
- Finalize Lessons Learned section
- Mark all repo items as FINAL (tag release version)
- Prepare device for live demo
- Submit & present final project